# Frontiers of Large Language Models in Medical Report Generation: Foundations, Methods, and Future Perspectives

Pengchen Liang, Jianguo Chen, *Member, IEEE*, Xiangwei Gong, Haishan Huang, Bin Pu

**Large language models (LLMs) are reshaping the landscape of clinical documentation by enabling automated, high-quality medical report generation. This survey provides an in-depth exploration of the methodologies, limitations, and future opportunities in applying LLMs to medical reporting tasks. We review foundational techniques across various clinical domains—including radiology, dermatology, mental health, and neurology—focusing on three critical dimensions: data preprocessing and feature extraction leveraging natural language processing (NLP) and computer vision techniques; model development through domain-adaptive pretraining and transfer learning strategies; and report generation refinement aimed at enhancing diagnostic reliability while maintaining computational efficiency. Comparative evaluations highlight that domain-specific medical LLMs consistently outperform general-purpose counterparts, with retrieval-augmented generation (RAG) models achieving an accuracy of 91.4%, surpassing human-authored reports at 86.3%, while reducing turnaround time from approximately 10 minutes to 15–20 seconds. Despite these promising results, unresolved challenges remain, including evaluation biases, risks of factual hallucination, and patient data privacy concerns. Looking ahead, we identify promising avenues such as integration with three-dimensional imaging, multimodal data fusion, and seamless incorporation into real-world healthcare workflows, paving the way for safer, more reliable, and scalable AI-assisted medical reporting.**

***Index Terms*—Large language models, medical report generation, artificial intelligence, natural language processing, healthcare automation, clinical decision support**

## I. Introduction

The healthcare industry is experiencing a paradigm shift driven by the rapid advancement of artificial intelligence (AI) technologies. AI aims to mimic human cognitive functions and is bringing transformative changes to healthcare, powered by the increasing availability of healthcare data and rapid progress in analytics techniques [1]. The integration of AI into medical practice has demonstrated significant potential across various domains, from diagnostics and treatment planning to patient monitoring and clinical decision support [2].

This work is partially funded by the National Natural Science Foundation of China under Grant 62372486, the Natural Science Foundation of Guangdong Province under Grant 2023A1515011179.

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Among the most promising developments is the emergence of large language models (LLMs), which have revolutionized natural language processing and demonstrated exceptional performance across diverse medical specialties, including radiology, nephrology, and dermatology [3]. These models exhibit the remarkable ability to provide proficient responses to free-text queries while demonstrating nuanced understanding of professional medical knowledge [4]. Recent advances in LLMs, particularly models like GPT-4 and specialized medical LLMs, have demonstrated capabilities in clinical language understanding tasks, including named entity recognition, relation extraction, natural language inference, and medical question-answering [4].

Medical imaging plays a crucial role in clinical practice for diagnosis and treatment, generating vast amounts of visual data that require expert interpretation and comprehensive reporting. The process of medical report generation is fundamental to healthcare delivery, serving as the primary means of communicating diagnostic findings, treatment recommendations, and patient status among healthcare providers. However, traditional report-writing processes present significant challenges: they can be error-prone for inexperienced physicians and time-consuming and tedious for experienced practitioners [5]. The manual generation of medical reports often involves complex reasoning processes, requiring clinicians to integrate visual patterns with informative linguistic descriptions while maintaining clinical accuracy and completeness.

Recent advances in vision-language models (VLMs) have opened new possibilities for addressing these challenges. VLMs combine computer vision and natural language processing to analyze visual and textual medical data, offering promising solutions for medical report generation and visual question answering [6]. These models leverage Transformer-based architectures to enable effective learning from multimodal data, facilitating the automated generation of structured, coherent, and clinically relevant reports from medical images. Hybrid retrieval-generation approaches have also emerged, reconciling traditional retrieval-based methods with modern learning-based approaches to achieve structured, robust, and diverse report generation [7].

The application of AI in specialized medical domains has shown remarkable progress. In dermatology, AI-aided diagnosis based on image recognition has become a modern focus, with 3D imaging systems and dermatoscopes combined with intelligent software helping clinicians screen and label skin lesions more effectively [8]. The potential for LLMs to enhance clinical documentation, support decision-making, and improve healthcare workflow efficiency has generated considerable excitement within the medical community [3]. Despite

the promising advances in AI-based medical report generation, several critical challenges persist that limit the widespread adoption and clinical deployment of these technologies. These challenges span technical, ethical, and practical dimensions, creating significant barriers to real-world implementation.

**Evaluation and Assessment Limitations:** One of the most significant challenges is the lack of appropriate evaluation metrics that accurately capture medical correctness and clinical relevance [9]. Current evaluation approaches predominantly rely on traditional Natural Language Processing (NLP) metrics such as BLEU, ROUGE, and METEOR, which fail to adequately assess the medical accuracy and clinical utility of generated reports. These metrics primarily focus on lexical similarity rather than semantic correctness or clinical significance, leading to potential misalignment between automated evaluation scores and actual clinical value [10, 11]. Systematic reviews of LLM evaluations in clinical medicine have revealed significant gaps in reliability, generalizability, and applicability of current assessment methodologies [12].

**Technical Complexity and Coherence:** Medical reports typically require the generation of long, coherent texts that maintain consistency across multiple sentences and paragraphs. This presents unique challenges for current AI models, which may struggle with maintaining coherence and clinical logic throughout extended narratives [5]. The hierarchical nature of medical reports, which often include sections for findings, impressions, and recommendations, requires sophisticated architectures capable of managing complex document structures. While recent approaches have attempted to address these challenges through hierarchical LSTM models and advanced architectures like Dia-LLaMA [13], ensuring consistency and clinical accuracy across extended texts remains an ongoing challenge.

**Ethical and Legal Concerns:** The deployment of LLMs in healthcare raises significant ethical and legal concerns that must be carefully addressed before widespread clinical adoption [14]. Key ethical issues include bias, plagiarism, copyright concerns, transparency, legal liability, and the risk of generating incorrect or misleading medical information. The high-stakes nature of medical decision-making amplifies these concerns, as errors in AI-generated reports could have serious consequences for patient safety and care outcomes [15, 16]. The imperative for regulatory oversight of LLMs in healthcare has become increasingly apparent, with calls for careful regulation that ensures safety and maintains ethical standards without stifling innovation [17].

**Explainability and Trust:** The black-box nature of many AI models poses significant challenges for clinical adoption, where understanding the reasoning behind diagnostic or therapeutic recommendations is crucial for maintaining physician trust and patient safety [18]. Explainable AI (XAI) has emerged as a critical requirement for medical applications, with the need for transparency becoming increasingly important as AI systems are integrated into clinical decision-making processes [19]. Current challenges include the need for interpretable explanations that can be understood by healthcare professionals, the balance between model accuracy and explainability, and the development of user-centric explanation interfaces [20, 21].

**Privacy and Data Security:** The use of sensitive medical data in training and deploying LLMs raises significant privacy concerns that must be addressed to ensure patient confidentiality and comply with healthcare regulations [22]. Medical data often contains highly sensitive personal information, and the potential for data leakage or misuse poses serious risks to patient privacy and institutional liability. While recent advances in privacy-preserving AI approaches, including federated learning and local deployment strategies, offer promising solutions [23, 24], comprehensive approaches to privacy protection remain a critical research priority [25].

Given the complexity and multifaceted nature of challenges in LLM-based medical report generation, there is an urgent need for a comprehensive survey that systematically analyzes the current state of the field, identifies key challenges, and provides guidance for future research directions. This survey aims to fill this gap by providing a systematic analysis of the current state, challenges, and future directions in large language model-based medical report generation.

The primary contributions of this comprehensive survey include:

1) **Systematic Analysis of Current Approaches:** We provide a comprehensive analysis of current LLM approaches for medical report generation, examining their architectures, training strategies, performance characteristics, and clinical applications across different medical domains.
2) **Challenge Assessment and Categorization:** We systematically evaluate existing challenges and limitations, including assessment metrics, technical complexity, ethical considerations, explainability requirements, and privacy concerns, providing a structured framework for understanding the barriers to clinical deployment.
3) **Solution Landscape Review:** We examine recent advances in addressing these challenges, including privacy-preserving approaches, explainable AI methods, improved evaluation frameworks, and innovative architectural designs that enhance model performance and clinical applicability.
4) **Future Research Roadmap:** We identify future research directions and provide actionable recommendations for advancing the field toward safe, effective, and ethical clinical deployment of LLM-based medical report generation systems.

Recent developments in the field have demonstrated both the potential and the complexity of this research area. Studies have shown the potential for iterative retrieval-augmented generation approaches to improve medical question-answering capabilities [26], while others have demonstrated how optimization of clinical guidelines interpretation can be achieved through structured formatting and advanced prompt engineering [27]. The integration of transformer models in healthcare presents both significant opportunities and notable risks, requiring careful consideration of benefits, shortcomings, and implementation challenges [28].

This survey aims to provide researchers, clinicians, and policymakers with a comprehensive understanding of the current

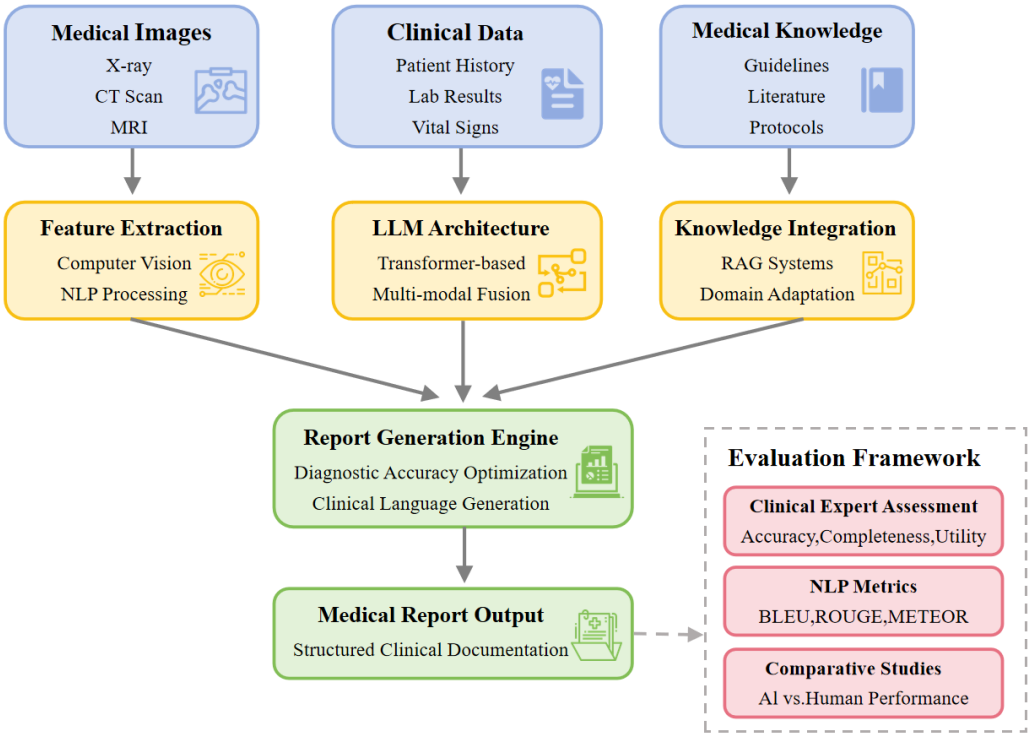



Fig. 1: System architecture of the LLM-based medical report generation framework, showing the integration of multimodal inputs (medical images, clinical data, and medical knowledge), feature extraction, transformer-based fusion, knowledge retrieval and adaptation, the report generation engine, and the evaluation framework.

landscape and informed guidance for future development and deployment decisions in this rapidly evolving field (As shown in the Figure 1). By systematically analyzing the state-of-the-art, identifying key challenges, and proposing future research directions, we hope to contribute to the responsible and effective advancement of LLM-based medical report generation technologies that can ultimately improve healthcare delivery and patient outcomes.

## II. Large Language Model Technical Foundation

### A. Concepts and Development of Large Language Models

Large Language Models (LLMs) represent a revolutionary advancement in artificial intelligence, characterized by their ability to understand, generate, and manipulate human language with unprecedented sophistication. These models are built on transformer architectures and trained on vast amounts of text data, enabling them to develop a comprehensive understanding of language patterns, semantic relationships, and contextual dependencies [29]. The development of LLMs has marked a paradigm shift from traditional rule-based and statistical approaches to more sophisticated neural network architectures capable of human-like language processing.

The evolution of LLMs in healthcare has been particularly noteworthy, with generative pretrained transformer models demonstrating enhanced capabilities and performance compared to existing AI models [30]. These models excel in natural language processing (NLP), enhancing clinical support, diagnosis, treatment, and medical research [31]. Breakthroughs, like GPT-4 and BERT (Bidirectional Encoder Representations from Transformer), demonstrate LLMs' evolution through improved computing power and data [31]. Unlike traditional AI systems that require extensive training data, LLMs can perform effectively with limited domain-specific training, making them particularly valuable for medical applications where specialized datasets may be scarce. This capability has enabled rapid adaptation of general-purpose models like GPT-3 and GPT-4 to specific healthcare domains through techniques such as prompt engineering and fine-tuning.

Recent developments in 2024-2025 have shown remarkable advances in LLM capabilities. Large language models (LLMs) are rapidly advancing medical artificial intelligence, offering revolutionary changes in health care [31]. Studies have demonstrated that open-source LLMs like OpenBioLLM-Llama3-70B outperform top models like GPT-4 and Med-PaLM, indicating the rapid democratization of advanced language modeling capabilities. The landscape has evolved to include over 20 specialized healthcare LLMs, each optimized for specific medical applications ranging from clinical decision support to biomedical research.

### B. Transformer Architecture and Technical Foundations

The technical foundation of LLMs relies on the transformer architecture, which utilizes attention mechanisms to capture long-range dependencies in text sequences. The architecture of the transformer builds the backbone of large language models (LLM), enabling them to harness the power of vast amounts of

data to gain a more profound understanding of the underlying information [32]. This architecture enables the models to understand complex medical terminology, clinical reasoning patterns, and the intricate relationships between symptoms, diagnoses, and treatments. The attention mechanism allows the model to focus on relevant parts of the input when generating responses, making it particularly suitable for medical report generation where specific clinical findings must be accurately correlated with appropriate diagnostic conclusions [29].

The success of transformer-based models can be attributed to several key factors: the self-attention mechanism, integrated encoder-decoder architecture, and scalable modular structure [32]. These characteristics allow it to learn effective representations of the underlying data, encode long-range dependencies, and process huge amounts of data in an efficient way. The basic building block of the transformer is the self-attention mechanism, which allows the model to learn complex sequence representations by incorporating or attending to information throughout other parts of the same sequence [32].

Recent architectural innovations have emerged to address computational efficiency challenges. Attention contributes by propagating information across time, unlike the MLP block, which solely performs channel mixing. New approaches include Low-Rank Adaptation (LoRA), parameter-efficient fine-tuning methods, and quantization techniques that reduce computational requirements while maintaining performance. QLoRA utilizes quantization techniques to convert conventional 16-bit pre-trained LLMs into 8-bit or 4-bit low-precision models, maintaining performance without the significant degradation [33].

### *C. Medical Domain Adaptation and Specialized Models*

The adaptation of general-purpose LLMs to medical domains has seen significant progress in 2024-2025. Continual pretraining on medical data to help the model better identify medical language by exposing it to clinical notes and biomedical literature (like PubMed) has become a standard approach for domain adaptation. Specialized models such as BioBERT, ClinicalBERT, and newer variants like Bio-ClinicalBERT have demonstrated superior performance on biomedical tasks [34]. BioBERT obtains higher F1 scores in biomedical NER (0.62) and biomedical RE (2.80), and a higher MRR score (12.24) in biomedical QA than the current state-of-the-art models [35]. The development process typically involves initializing with general language model weights, followed by continual pretraining on medical corpora, and finally fine-tuning for specific downstream tasks [32].

Recent research has also explored multimodal capabilities in medical LLMs. Multimodal LLMs are unique in processing multimodal medical data, thereby improving emergency, elder care, and digital medical procedures [31]. These models can integrate text, images, audio, and other data modalities, enabling more comprehensive medical analysis and decision-making support [36]. Recent studies have extensively evaluated LLM performance on medical licensing examinations, providing insights into their clinical readiness. GPT-4 scored 163/163 (100%), GPT-3.5 scored 134/163 (82.2%), Bard scored 123/163 (75.5%), and Claude scored 138/163 (84.7%) on National Board of Medical Examiners sample questions [37]. These results demonstrate substantial improvements over earlier models and approach or exceed human performance levels [38].

In USMLE step 1, step 2CK and Step 3, there are accuracy rates of 91.5%, 94.2% and 92.7% provided from GPT-4o, 93.2%, 95.0% and 92.0% provided from GPT-4 [39]. Studies across different cultural and linguistic contexts have shown varying performance levels, highlighting the importance of cultural adaptation and multilingual training for global healthcare applications [40, 41]. The sophistication of these models extends beyond simple text generation to include complex reasoning, clinical decision-making support, and the ability to synthesize information from multiple sources to provide comprehensive medical insights [29]. The MedExamLLM platform comprises data from 198 medical exams conducted in 28 countries, covering 15 languages, and spanning from 2009 to 2023, demonstrating the global scope of LLM evaluation in healthcare [39].

Recent developments in LLM training have focused on efficiency and specialized medical knowledge integration. Fine-tune an LLM on domain-specific corpora (e.g., medical literature) using unsupervised objectives like masked language modeling has become a standard approach. Parameter-efficient fine-tuning methods, including LoRA and QLoRA, have enabled smaller medical institutions to develop specialized models without extensive computational resources [42]. QLoRA algorithm provides a more memory- and computation-efficient fine-tuning solution compared to standard LoRA [33]. This has democratized access to advanced LLM capabilities, allowing hospitals and medical research institutions to develop specialized models tailored to their specific needs and patient populations [43].

The integration of knowledge graphs and external medical databases has also emerged as a crucial technique for enhancing LLM performance. Retrieval Augmented Generation (RAG) approaches allow models to access up-to-date medical information and evidence-based guidelines during inference, addressing concerns about knowledge cutoffs and outdated information. Despite remarkable progress, several challenges remain in the deployment of LLMs for healthcare applications. Current state-of-the-art LLMs do not accurately diagnose patients across all pathologies (performing significantly worse than physicians), follow neither diagnostic nor treatment guidelines, and cannot interpret laboratory results [44]. These limitations highlight the need for continued research and careful validation before widespread clinical deployment [45].

Fine-tuning LLMs with medical data can expose the training data used during learning to adversarial attacks. The success rate of attacks that caused the model to expose part of the training data was up to 21.8% [46]. Privacy and security concerns represent significant barriers to adoption, particularly given the sensitive nature of medical data [43]. Bias and fairness issues also persist, with studies showing that LLMs may exhibit differential performance across demographic groups and clinical contexts [31]. Critical challenges remain, including the need for standardized evaluation frameworks, attention to

ethical considerations, and the underrepresentation of high-priority medical specialties [47].

The field is rapidly evolving toward more sophisticated architectures and applications. Task specialization and fine-tuning: LLMs are now being fine-tuned for specific domains, such as healthcare (e.g., Med-PaLM 2), law, and science. Integration beyond text toward multimodal capabilities represents a significant trend, with models capable of processing images, audio, and video alongside textual information [36]. Safety mechanisms and ethical AI design principles are becoming increasingly important. Leading LLMs are now designed with improved safety protocols to minimize biased outputs. The development of robust evaluation frameworks and governance structures will be essential for responsible deployment in healthcare settings [43]. The integration of advanced explainable AI (XAI) techniques is also gaining prominence, addressing the critical need for transparency in medical AI applications [48]. By being able to explain and understand the predictions through XAI techniques, trust and confidence can be built in biomedical AI models, which is even more relevant for decision-making processes in the clinical domain [32].

The evolution of Large Language Models represents a paradigm shift in medical artificial intelligence, offering unprecedented capabilities for natural language understanding and generation in healthcare contexts [3]. From their foundational transformer architectures to specialized medical adaptations, these models have demonstrated remarkable progress in clinical benchmarks and real-world applications [49, 50]. However, significant challenges remain in ensuring accuracy, safety, privacy, and fairness for clinical deployment [51]. The continued advancement of LLMs, coupled with emerging techniques in multimodal processing, parameter-efficient training, and explainable AI, positions these technologies to transform healthcare delivery while requiring careful attention to ethical and safety considerations [31].

### *D. Clinical Specialties Applications*

The application of LLMs across various medical specialties has demonstrated their versatility and potential to transform healthcare delivery. Each specialty presents unique challenges and opportunities for LLM implementation, requiring specialized approaches and domain-specific adaptations (As shown in the Figure 2).

#### *1) Radiology and Medical Imaging*

The field of radiology has been at the forefront of LLM implementation in healthcare, with significant developments in automated medical report generation and diagnostic support systems. Advanced frameworks like PromptMRG have been developed to improve diagnostic accuracy through diagnosis-driven prompts, utilizing encoder-decoder architectures with disease classification branches to guide report generation [52]. These systems achieve state-of-the-art clinical efficacy performance by incorporating cross-modal feature enhancement and adaptive logit-adjusted loss to address disease imbalance issues. LLMs have demonstrated particular promise in translating complex radiology reports into plain language for patients and healthcare providers. Studies utilizing ChatGPT for translating radiology reports have shown successful translation capabilities with average scores of 4.27 on a five-point system, with minimal information loss and misinformation [53]. The comparison between ChatGPT and GPT-4 has revealed significant improvements in translation quality with newer model versions, suggesting rapid advancement in the field.

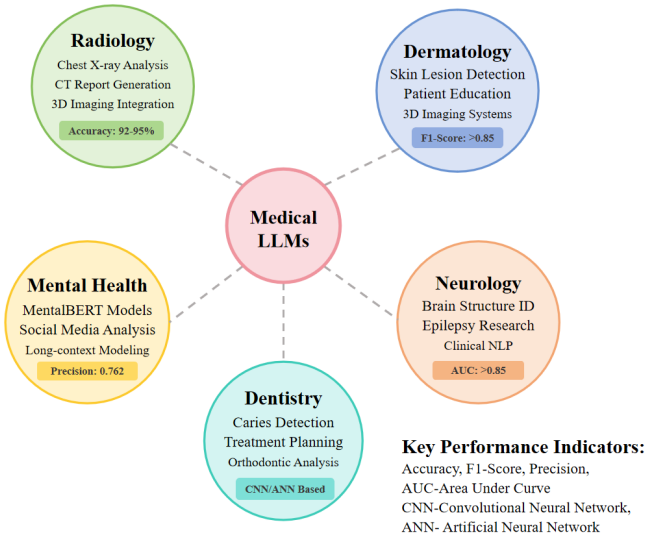



Fig. 2: Application domains and key performance indicators of medical LLMs across various specialties—Radiology, Dermatology, Neurology, Dentistry, and Mental Health—highlighting typical tasks and reported metrics (e.g. accuracy, F1-score, AUC, precision).

The emergence of multimodal large language models (MLLMs) has marked a paradigmatic shift in medical imaging applications. BrainGPT, a clinically visual instruction-tuned (CVIT) model designed for 3D CT radiology report generation, has demonstrated exceptional performance with an average FORTE F1-score of 0.71 and achieved 74% human-indistinguishable reports in Turing-like evaluations [54]. This development represents significant progress in extending MLLM capabilities beyond 2D imaging to complex 3D medical volumes. Vision-language models have shown remarkable capabilities in medical imaging tasks, with models like GPT-4V and Gemini Ultra demonstrating promising results in diagnosing complex pathological conditions. Recent evaluation studies of 15 open-source LLMs and GPT-4o across 1,933 Eurorad cases revealed that while GPT-4o maintained superior performance, open-source models like Llama-3-70B are rapidly closing the performance gap [55]. These findings suggest the democratization of advanced medical AI capabilities through accessible open-source alternatives.

Recent advances in automated radiology report generation have leveraged sophisticated fine-tuning approaches on domain-specific datasets. Studies utilizing fine-tuned Llama 3-8B models trained on over 21,000 radiology reports have demonstrated significant improvements in generating accurate and concise conclusions for MRI and CT examinations [56]. These systems achieve comparable performance to human-generated reports while substantially reducing ra-

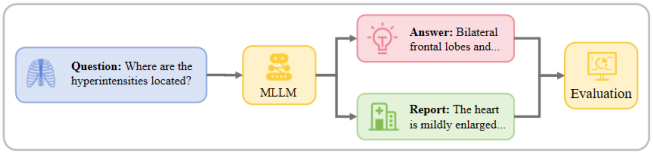



Fig. 3: Multimodal large language models in radiology.

diologist workload and improving reporting consistency. The development of specialized radiology LLMs, such as MGH Radiology Llama, represents a significant advancement in domain-specific medical AI. Built upon Llama 3 70B and trained on over 6.5 million de-identified medical reports from Massachusetts General Hospital, these models demonstrate superior performance in generating clinically relevant radiology impressions across various imaging modalities [57]. Such specialized models address the limitations of general-purpose LLMs by incorporating domain-specific knowledge and maintaining patient privacy through on-premise deployment capabilities.

The integration of LLMs with 3D medical imaging has opened new possibilities for enhanced diagnostic capabilities. Figure 3 illustrates a hypothetical workflow for a radiology multimodality LLM using CT as part of the input. DiffusionBlend, a novel framework for 3D CT reconstruction, demonstrates how generative AI can reconstruct high-quality sparse-view 3D CT scans with dramatically reduced radiation exposure while maintaining diagnostic quality [58]. This advancement addresses critical concerns about radiation exposure in medical imaging while preserving the diagnostic utility of imaging studies. Multimodal vision-language models have shown particular strength in pathology imaging applications. PathChat, a multimodal generative AI copilot for human pathology, employs specialized architectures combining vision transformers with large language models to provide comprehensive pathology analysis capabilities [59]. The CONCH model for histopathology analysis demonstrates zero-shot cross-modal retrieval capabilities, enabling automated analysis of pathology slides with confidence scoring for various diagnostic questions.

Despite promising technical achievements, the clinical implementation of LLMs in radiology faces significant challenges. Comprehensive systematic reviews of ChatGPT performance in radiology applications reveal that while 84.1% of studies demonstrate high performance, concerns remain regarding accuracy consistency across different imaging modalities and clinical scenarios [60]. Studies indicate accuracy rates varying from 81% for complex CT examinations to 98% for standardized mammography reports, highlighting the need for modality-specific model optimization. Clinical decision support applications of LLMs in radiology require careful consideration of workflow integration and user acceptance. Qualitative studies of AI-based diagnostic decision support system implementation reveal that clinicians prefer tool-based interactions for well-defined tasks rather than conversational interfaces, particularly under time pressure [61]. These findings emphasize the importance of designing AI tools that complement existing clinical workflows rather than requiring fundamental changes to established practices.

*2) Dentistry and Oral Health*

The integration of LLMs into dentistry has grown rapidly, with 17 studies (out of 4,079 records) focusing on ChatGPT for post-operative queries and diagnostic assistance, underscoring both broad applicability across specialties and the need to evaluate diverse LLM platforms [62]. Diagnostic evaluations on 1,490 NBDE multiple-choice items revealed Copilot (85.5%), Claude (84.0 %), and ChatGPT (83.8 %) leading performance, yet GPT-4 attained only 73 % integrated accuracy—below the clinical threshold—and significantly outperforming GPT-3.5 (54 %) and Bard (56 %) [63, 64]. In endodontics, ChatGPT-4o matched dental student accuracy on radiographic cases but struggled with artifacts and complex imaging factors [65].

Educational applications show mixed results: ChatGPT underperformed in periodontal surgery exams compared to students but effectively generated personalized feedback and case scenarios for charting and patient management [66, 67]. In patient education, ChatGPT delivered reliable preventive dentistry information (4.3/5) but less accurate content for oral surgery and cancer topics, highlighting the need for domain-specific training [68, 69]. Key technical challenges include developing multimodal foundation models capable of integrating radiographs, clinical photos, and histories for automated and cross-modal diagnosis [70]. Safety concerns—such as hallucinations, bias, and privacy risks—necessitate human oversight, treating LLMs as adjunct tools rather than standalone systems [68]. Future priorities involve standardized reporting, advanced prompting, dataset fine-tuning, practitioner training, and ethical guidelines to ensure responsible LLM deployment in dental practice [71, 72].

*3) Dermatology*

Dermatology—highly visual and dependent on clinical images, dermoscopy, and EHRs—has become a prime candidate for multimodal AI, especially LLMs and VLMs. Skin diseases affect 30–70% of people worldwide and rank fourth in nonfatal disease burden, while dermatologist shortages (approx. 3.65 per 100,000 in the US) limit access in underserved areas [73, 74]. Early AI efforts focused on single tasks (e.g., melanoma detection) but struggled with multimodal data integration [75].

Recent foundation models mark a paradigm shift. SkinGPT-4 aligns a frozen ViT encoder and Q-Former with Llama-2-13b-chat via a two-step strategy using 52,929 images. It generates natural-language diagnostic reports—feature identification, categorization, etiology, and treatment—and achieved 80.63% correct/relevant diagnoses, with 82.5% of etiologies and 85.6% of treatments deemed informative or useful by dermatologists [74]. PanDerm, trained on over 2 million images across four modalities (TBP, clinical, dermoscopic, dermatopathology), uses masked latent modeling and CLIP alignment to reach state-of-the-art results on 28 benchmarks—often outperforming prior methods with only 10% labeled data and 200 epochs versus 500–800 [75]. “Any-to-any” VLMs like Qwen 2.5 Omni’s Thinker-Talker incorporate Mixture of Experts, masked image modeling, and contrastive learning for efficient, scalable multimodal understanding [76]. Persistent

challenges include skin-tone bias, limited validation in diverse populations, and privacy. SkinGPT-4's offline deployment mode addresses HIPAA concerns by processing data locally without retention [74].

*4) Mental Health and Psychiatry*

The global rise in mental health disorders—exacerbated by treatment gaps affecting over 75% of individuals in low- and middle-income countries—and the shortage of professionals have spurred AI solutions, with 40 studies exploring LLMs for screening, interventions, and clinical decision support [77, 78]. Specialized models such as Mental-Alpaca and Med-PaLM 2 achieved 77.5% accuracy on DSM-5 diagnostic cases (rising to 92.5% for broader categories), while pilot inpatient studies reported improved WHO Quality of Life scores and high user satisfaction with ChatGPT–facilitated sessions [77, 79]. Nevertheless, qualitative reviews highlight inadequate professionalism, superficial assessments, and mixed usability compared to human therapists, with 40% of psychiatrists open to LLM use yet concerned about reliability and confidentiality [78].

Rapid growth of AI companions (e.g., Woebot, ChatGPT repurposed) has raised safety alarms—cases of dependency, crises, and litigation (Character.AI lawsuit after teen suicide), and reports of self-harm encouragement by some bots [80]. Technically, current LLMs lack depth for clinical assessment, risk oversimplification, and operate largely outside HIPAA protections [81, 82]. Experts advise LLMs as adjuncts under human oversight, supporting psychoeducation and structured interventions while emphasizing bias mitigation, privacy safeguards, and ethical frameworks co-developed with clinicians and diverse communities [51, 83, 84].

*5) Diabetes Care*

LLMs have shown significant potential in diabetes management, with studies demonstrating their competence in diabetes-related queries across multiple languages [85]. Evaluation of ten different LLMs on diabetes certification examinations has revealed that models like ChatGPT-4.0 can achieve passing scores and effectively assist primary care physicians in diabetes training and examination performance. The role of LLMs in diabetes care extends beyond education to clinical decision-making support. Studies comparing LLM-driven decisions with those made by experienced endocrinologists have highlighted both the potential contributions and limitations of these models in supporting diabetes care [86]. The complexity of diabetes management, with its multiple guideline-concordant treatment options and patient-specific considerations, presents both opportunities and challenges for LLM implementation.

Machine learning frameworks for diabetes prediction and diagnosis have achieved remarkable accuracy rates, with some models achieving precision, sensitivity, and F1-scores exceeding 97% [87]. These frameworks incorporate advanced preprocessing techniques and optimized classification models, demonstrating the potential for AI to significantly improve diabetes care delivery. The potential of LLMs in diabetes care encompasses various applications, from early diagnosis and timely interventions to continuous self-care support and patient empowerment [88]. The integration of LLMs with digital health technologies presents opportunities to address resource constraints and improve accessibility of diabetes care globally.

*6) Neurology and Brain Sciences*

The integration of AI in neurology has revolutionized diagnosis, treatment, and research approaches for neurological disorders [89]. AI algorithms can accurately identify brain structures and detect abnormalities through automated image interpretation, accelerating diagnosis while reducing workload on medical professionals. These systems can perform sophisticated perceptual and cognitive tasks, including object identification and decision-making, that are crucial for neurological assessment. Natural language processing applications in epilepsy research have demonstrated significant potential for transforming clinical care [90]. Studies have shown that 58% of NLP applications in epilepsy focus on classifying clinical records into predefined categories, improving patient identification and treatment decisions. Other applications include structured clinical information retrieval from electronic health records, scientific papers, and patient-generated content.

The evaluation of ChatGPT in specialized neurosurgical knowledge has revealed both capabilities and limitations [91]. While the model can generate informative content about emerging procedures like middle meningeal artery embolization, concerns remain about the currency and accuracy of information, particularly for rapidly evolving surgical techniques. Natural language processing in clinical neuroscience and psychiatry has shown high performance, with average F1-scores and AUC values above 85% [92]. However, challenges include strong asymmetry between English and non-English models, difficulty obtaining high-quality annotated data, and training biases that affect generalizability.

### *E. Healthcare System Integration*

The integration of LLMs into healthcare systems represents a fundamental shift toward value-based treatment and personalized care delivery [93]. As shown in Figure 4, modern healthcare equipment and devices now integrate internal applications that collect and store comprehensive patient data, providing rich resources for ML-driven predictive models. This integration enables healthcare providers to deliver personalized and highly effective care while optimizing resource allocation.

*1) Implementation Considerations*

The operationalization of LLMs in healthcare systems requires careful consideration of multiple technical and practical factors [30]. Key implementation considerations include processing needs and information systems infrastructure, operating costs, model biases, and evaluation metrics. Healthcare organizations must also address Health Insurance Portability and Accountability Act (HIPAA) compliance, build trust with healthcare providers, and establish broader access to LLM tools.

The computational requirements for LLMs pose significant challenges for healthcare organizations, requiring substantial processing power and storage capabilities. However, advances in model optimization and cloud-based deployment have made these technologies more accessible to organizations of varying sizes and technical capabilities.

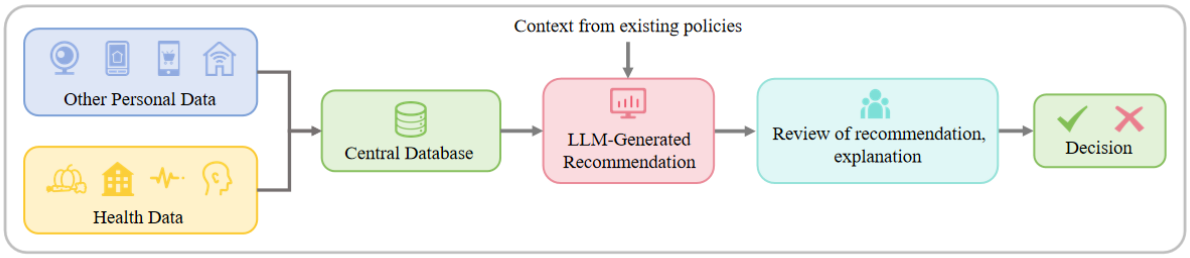



Fig. 4: Hypothetical workflow of a clinical credit system involving multimodal data.

*2) Sentiment Analysis and Patient Communication*

LLMs have demonstrated superior performance in sentiment analysis of health-related survey data compared to traditional tools [94]. Studies comparing ChatGPT and fine-tuned open pre-trained transformers against widely used sentiment analysis tools have shown that LLMs, particularly ChatGPT, outperform conventional approaches with higher accuracy and F-measure scores. This capability is particularly valuable for analyzing patient feedback, monitoring treatment responses, and understanding patient experiences. The application of LLMs in patient portal communication has shown promising results for emergency triage [95]. Knowledge graph-based retrieval-augmented generation approaches have achieved high accuracy (0.99), sensitivity (0.98), and specificity (0.99) in triaging emergency patient messages, demonstrating the potential for automated patient communication systems.

*3) Personalization and Adaptation*

LLMs have demonstrated remarkable plasticity in adapting to specific personality structures and psychopathologies [96]. Studies have shown that ChatGPT can generate mentalizing-like responses consistent with different psychological conditions, accurately describing emotional reactions of individuals with various personality disorders. This capability suggests potential for personalized healthcare communication and intervention strategies. The ability of LLMs to understand and adapt to individual patient characteristics extends beyond psychology to encompass various aspects of personalized medicine. This includes adapting communication styles, treatment recommendations, and educational materials to match patient preferences, literacy levels, and cultural backgrounds.

### *F. Technical Challenges and Limitations*

Despite the significant advances in LLM technology for healthcare applications, several technical challenges and limitations persist that must be addressed for successful clinical implementation.

*1) Model Performance and Reliability*

While LLMs have shown impressive performance in various healthcare tasks, concerns remain about their reliability and consistency in clinical settings. The probabilistic nature of these models means that they can produce different outputs for identical inputs, which poses challenges for clinical decision-making where consistency is crucial. Additionally, the potential for "hallucinations", which may generate plausible but factually incorrect information, remains a significant concern in the medical applications.

*2) Domain Adaptation and Specialization*

The adaptation of general-purpose LLMs to specific medical domains requires substantial effort and expertise. While techniques like fine-tuning and prompt engineering have shown promise, the development of truly specialized medical LLMs often requires domain-specific datasets, expert knowledge, extensive validation, ethical compliance, privacy, and interpretability. The challenge is particularly acute for rare diseases or emerging medical conditions where training data may be limited.

*3) Computational Requirements and Scalability*

The computational demands of LLMs present significant challenges for healthcare organizations, particularly smaller institutions with limited IT infrastructure. The trade-off between model performance and computational efficiency remains an ongoing concern, with researchers working to develop more efficient architectures and deployment strategies that can deliver high-quality results with reduced computational overhead. This technical foundation provides the basis for understanding how LLMs can be effectively applied to medical report generation, setting the stage for more detailed examination of specific applications, challenges, and future directions in subsequent sections of this survey.

## III. Large Language Model-Based Medical Report Generation Methods

### *A. Data Preprocessing and Feature Extraction*

The effectiveness of large language model-based medical report generation systems fundamentally depends on robust data preprocessing and feature extraction techniques. This section examines the key methodologies employed to prepare and process medical data for optimal model performance, encompassing natural language processing techniques, unsupervised learning approaches, and computer vision methods.

*1) Natural Language Processing*

Natural language processing serves as the foundation for understanding and processing medical text data in report generation systems. The integration of NLP techniques with healthcare applications has led to transformative improvements in conversational AI and medical text understanding [97]. Modern approaches leverage sophisticated NLP frameworks that facilitate seamless healthcare conversations and enable

comprehensive analysis of medical documentation. Domain-specific language model development has emerged as a critical strategy for capturing the nuances of medical language. Research has demonstrated the importance of domain-specific continued pretraining for capturing long contextual information, particularly in specialized areas such as mental health [98]. The development of specialized models like MentalXLNet and MentalLongformer has shown that domain-specific pretraining significantly improves performance on tasks requiring understanding of extended medical narratives and complex clinical relationships.

The preprocessing of medical text involves several sophisticated techniques to handle the unique characteristics of clinical documentation. Medical texts often contain specialized terminology, abbreviated forms, and domain-specific linguistic patterns that require careful handling during preprocessing. Tokenization strategies must account for medical acronyms, drug names, and clinical measurement units, while maintaining semantic coherence across different medical specialties. Text normalization in medical NLP involves standardizing medical terminology, handling negation and uncertainty expressions, and resolving temporal references that are crucial for accurate clinical understanding. Advanced preprocessing pipelines incorporate medical ontologies and knowledge bases to ensure consistent interpretation of medical concepts and their relationships.

*2) Unsupervised Learning Models*

Unsupervised learning approaches have shown remarkable potential in medical report generation, particularly in addressing the challenge of limited paired data in healthcare domains. The development of Knowledge Graph Auto-Encoder (KGAE) represents a significant advancement in unsupervised medical report generation [99]. This approach accepts independent sets of images and reports during training, utilizing a pre-constructed knowledge graph as a shared latent space to bridge visual and textual domains. The KGAE framework consists of knowledge-driven encoders and decoders that project medical images and reports to corresponding coordinates in the latent knowledge space. This unsupervised approach eliminates the dependency on paired image-report datasets, which are often expensive and time-consuming to construct in medical domains. Experimental results demonstrate that KGAE generates desirable medical reports without requiring any image-report training pairs, while also achieving superior performance in semi-supervised and supervised settings.

Contrastive learning methods have also shown promise in unsupervised domain adaptation for medical imaging tasks. Research in OCT segmentation has demonstrated that semi-supervised learning frameworks combining supervised and contrastive learning can achieve significant improvements in cross-domain performance [100]. The introduction of contrastive pairing schemes that leverage similarity between nearby slices in 3D medical images, integrated with channel-wise aggregation for contrastive feature map projection, has led to substantial improvements in domain adaptation scenarios. Generative adversarial networks (GANs) have been successfully applied to unsupervised anomaly detection in medical imaging, particularly in mammography for breast cancer screening [101]. These approaches train generative models on normal medical images and perform anomaly detection to identify pathological conditions, demonstrating high sensitivity while overcoming the limitations of supervised methods that require extensive labeled datasets. Universal medical self-supervised learning frameworks like UniMiSS+ have advanced the field by enabling cross-dimensional learning from both 2D and 3D medical images [102]. These frameworks utilize pyramid U-like medical Transformers as backbones and incorporate digitally reconstructed radiographs (DRR) technology to create paired multimodal data, enabling more effective cross-modality correlation learning and improved medical image analysis performance.

*3) Computer Vision*

Computer vision techniques form a crucial component of medical report generation systems, particularly in vision-language models (VLMs) that combine visual analysis with textual generation. Medical VLMs integrate computer vision and natural language processing to analyze visual and textual medical data simultaneously [6]. These models utilize Transformer-based architectures to enable effective learning from multimodal data, facilitating the automated generation of structured, coherent, and clinically relevant reports from medical images. The integration of computer vision in medical report generation involves sophisticated image analysis techniques that can identify anatomical structures, detect abnormalities, and quantify pathological changes. Advanced preprocessing techniques include image normalization, contrast enhancement, and region-of-interest detection that prepare medical images for optimal analysis by deep learning models.

Recent developments in radiology image analysis have emphasized the importance of collaborative knowledge integration in generating accurate radiology findings [103]. Two-step approaches that first detect regions with abnormalities and then generate relevant text descriptions have shown superior performance compared to end-to-end image captioning methods. This approach introduces interpretability and aligns with the systematic reasoning that radiologists employ when reviewing medical images. The development of automated radiosynthesis and imaging protocols has also contributed to improved computer vision applications in medical report generation [104]. Consistent and reproducible protocols for clinical dose preparation and imaging procedures provide standardized inputs for computer vision systems, enabling more reliable and accurate automated analysis. Vision-language integration strategies involve multiple approaches for fusing visual and textual information. Early fusion methods combine image features with text embeddings at the input level, while late fusion approaches process visual and textual information separately before combining them at the decision level. Attention-based fusion mechanisms enable dynamic weighting of visual and textual features based on their relevance to specific reporting tasks.

### *B. Model Construction and Training*

The construction and training of large language models for medical report generation involves strategic decisions regard-

ing model architecture, pretraining approaches, and optimization techniques. This section examines both domain-specific pretraining from scratch and transfer learning approaches that leverage existing pretrained models.

*1) Domain-Specific Pretraining from Scratch*

Domain-specific pretraining has emerged as a powerful approach for developing specialized medical language models that outperform general-domain models on healthcare tasks. Research has demonstrated that for domains with abundant unlabeled text, such as biomedicine, pretraining language models from scratch can result in substantial gains over continual pretraining of general-domain models [105]. This approach challenges the prevailing assumption that domain-specific pretraining should start from general-domain language models. The development of biomedical language models like BioBERT and ClinicalBERT has established the foundation for domain-specific pretraining in healthcare. Comprehensive biomedical NLP benchmarks have shown that domain-specific pretraining serves as a solid foundation for a wide range of biomedical NLP tasks, leading to state-of-the-art results across various medical applications. These models demonstrate improved performance in tasks such as named entity recognition, relation extraction, and clinical text classification.

RadBERT represents a significant advancement in radiology-specific language model development [106]. This family of BERT-based language models was specifically adapted for radiology applications, with variants pretrained on millions of radiology reports from healthcare systems. The research demonstrated that transformer-based language models tailored to radiology achieve improved performance on radiology NLP tasks, including abnormal sentence classification, report coding, and report summarization, compared to baseline transformer language models. The pretraining process for domain-specific medical models involves careful curation of medical text corpora, consideration of medical vocabulary and terminology, and adaptation of pretraining objectives to medical contexts. Medical text pretraining requires handling of specialized medical language patterns, including clinical abbreviations, drug nomenclature, and complex medical relationships that may not be adequately represented in general-domain corpora. Architecture modifications for medical domain pretraining include adaptation of attention mechanisms to handle medical text characteristics, incorporation of medical knowledge graphs and ontologies, and modification of tokenization strategies to better handle medical terminology. These adaptations ensure that the resulting models can effectively capture the semantic richness and complexity of medical language.

*2) Transfer Learning from Pretrained Models*

Transfer learning approaches leverage existing pretrained models and adapt them to medical domains, offering a more resource-efficient alternative to training from scratch. The Linear Growth Operator (LiGO) framework represents an innovative approach to accelerating transformer training by learning to grow pretrained models [107]. This method learns to linearly map parameters from smaller models to initialize larger models, utilizing width- and depth-growth operators with Kronecker factorization to encode architectural knowledge. LiGO demonstrates significant computational efficiency gains, saving up to 50% of the computational cost compared to training from scratch while consistently outperforming baseline methods. This approach is particularly valuable in medical applications where computational resources may be limited and rapid model development is crucial for clinical deployment.

Pathology-specific applications have revealed nuanced insights about the impact of pretraining domain relevance on medical NLP tasks [108]. Research examining various deep learning-based NLP models on pathology reports has shown that the optimal pretraining strategy varies depending on the specific task requirements. Models pretrained on medical data demonstrate superior performance when medical domain knowledge is crucial, such as in CPT code classification. However, for more subjective interpretation tasks, models fine-tuned from general or unrelated text sources can achieve comparable or superior results. Multi-modal pretraining approaches have been developed specifically for medical vision-language understanding and generation [109]. The development of comprehensive radiographic datasets like RadioGraphy Captions (RGC) has enabled systematic investigation of key factors affecting medical VLP performance. These studies provide insights for making informed pretraining decisions and establish strong baselines for various medical vision-language tasks.

Instruction-tuned generative large language models have shown particular promise in medical applications when combined with domain-adaptive pretraining [110]. Systems that leverage large-scale medical text data for domain-adaptive pretraining of instruction-tuned LLMs demonstrate enhanced medical knowledge and improved performance on specific medical tasks, particularly in zero-shot settings for radiology report impression generation. The selection of base models for transfer learning involves consideration of model architecture, pretraining data characteristics, and task-specific requirements. Popular base models include BERT variants, RoBERTa, and more recent transformer architectures that offer different trade-offs between performance and computational efficiency.

### C. Report Generation and Strategy Optimization

The final stage of medical report generation involves implementing sophisticated strategies to optimize both the accuracy and efficiency of the generation process. This section examines advanced techniques for improving diagnostic accuracy and optimizing computational efficiency in medical report generation systems.

*1) Improving Diagnostic Accuracy in Medical Report Generation*

Enhancing diagnostic accuracy represents one of the most critical challenges in medical report generation, requiring sophisticated approaches that can handle the complexity and nuance of medical decision-making. The PromptMRG framework introduces diagnosis-driven prompts that significantly improve diagnostic accuracy through explicit guidance during the generation process [52]. This encoder-decoder architecture

incorporates an additional disease classification branch that converts diagnostic results into token prompts to guide report generation. PromptMRG addresses the challenge of imbalanced disease distributions through adaptive logit-adjusted loss functions applied to the classification branch. This approach considers the individual learning status of each disease, enabling more reliable diagnosis of rare diseases that are typically underrepresented in training data. The framework also incorporates cross-modal feature enhancement, utilizing pretrained CLIP models to retrieve similar reports from databases and assist in diagnosis through knowledge transfer.

Experimental validation on standard medical report generation benchmarks has demonstrated that PromptMRG achieves state-of-the-art clinical efficacy performance, with significant improvements in both common and rare disease identification. The framework's ability to handle disease imbalance while maintaining high diagnostic accuracy makes it particularly valuable for real-world clinical applications where rare conditions must be reliably detected. AutoMedPrompt represents another significant advancement in optimizing medical language models through textual gradient-based prompt optimization [111]. This framework leverages TextGrad's automatic differentiation capabilities to improve the medical reasoning abilities of general foundation LLMs without requiring extensive fine-tuning. The approach has demonstrated superior performance on multiple medical QA benchmarks, including MedQA, PubMedQA, and specialized nephrology assessments.

The AutoMedPrompt framework achieves state-of-the-art performance on PubMedQA with 82.6% accuracy while outperforming previous prompting strategies on open-source models. This approach is particularly significant because it enables general-purpose LLMs to achieve specialized medical performance through prompt optimization rather than domain-specific training, making advanced medical AI capabilities more accessible to healthcare organizations with limited resources. Advanced prompt engineering techniques for medical applications involve careful consideration of medical reasoning patterns, incorporation of clinical guidelines and protocols, and adaptation to specific medical specialties. These techniques enable more precise control over model behavior and can significantly improve performance on specialized medical tasks without requiring extensive model retraining.

*2) Computational Cost Optimization*

Computational efficiency optimization has become increasingly important as medical AI systems scale to handle larger datasets and more complex models. The Linear Growth Operator (LiGO) approach demonstrates how knowledge transfer from smaller to larger models can achieve substantial computational savings [107]. By learning linear mappings between model parameters, LiGO reduces training computational costs by up to 50% while maintaining or improving model performance. Retrieval Augmented Generation (RAG) systems represent another approach to computational optimization that enhances model capabilities without increasing model size [112]. Medical RAG systems combine the knowledge representation capabilities of large language models with external knowledge retrieval, enabling rapid access to specialized medical information during inference. Case studies in preoperative medicine have demonstrated that RAG-enhanced models can generate responses in 15-20 seconds compared to 10 minutes for human-generated responses, while achieving accuracy levels comparable to human experts.

The development of RAG systems for medical applications involves sophisticated vector storage techniques and embedding model optimization. These systems utilize frameworks like LangChain and Llamaindex for processing clinical documents, with Pinecone vector storage providing efficient retrieval capabilities. The optimization of embedding dimensionality and similarity metrics enables rapid and accurate retrieval of relevant medical information during report generation. RAG systems offer several advantages for medical applications, including grounded knowledge integration, upgradability without model retraining, and scalability across different medical domains. The ability to update knowledge bases without retraining large models is particularly valuable in rapidly evolving medical fields where new research and guidelines emerge frequently.

Efficient training strategies also include progressive training approaches where models are initially trained on smaller datasets or simplified tasks before advancing to more complex scenarios. This approach can reduce overall training time while improving final model performance by enabling more stable learning trajectories. Memory-efficient architectures and gradient accumulation techniques enable training of larger models with limited computational resources. These approaches are particularly important for healthcare organizations that may not have access to large-scale computing infrastructure but require high-performance medical AI capabilities. The optimization of inference efficiency involves techniques such as model quantization, knowledge distillation, and efficient attention mechanisms that reduce computational requirements during deployment. These optimizations are crucial for real-time medical applications where rapid response times are essential for clinical workflow integration. This comprehensive examination of medical report generation methods provides the foundation for understanding current capabilities and limitations, setting the stage for detailed analysis of challenges and future directions in subsequent sections of this survey.

## IV. Performance Evaluation

The assessment of large language model–based medical report generation systems requires comprehensive evaluation frameworks that can accurately measure both technical performance and clinical utility. This section examines the critical evaluation methodologies employed in the field, focusing on comparative studies between AI systems and human medical professionals, as well as multi-dimensional evaluation approaches that combine clinical expert assessment with natural language processing metrics.

### *A. Comparative Studies: AI Models vs. Human Medical Professionals*

Comparative evaluation between AI systems and human medical professionals represents one of the most crucial

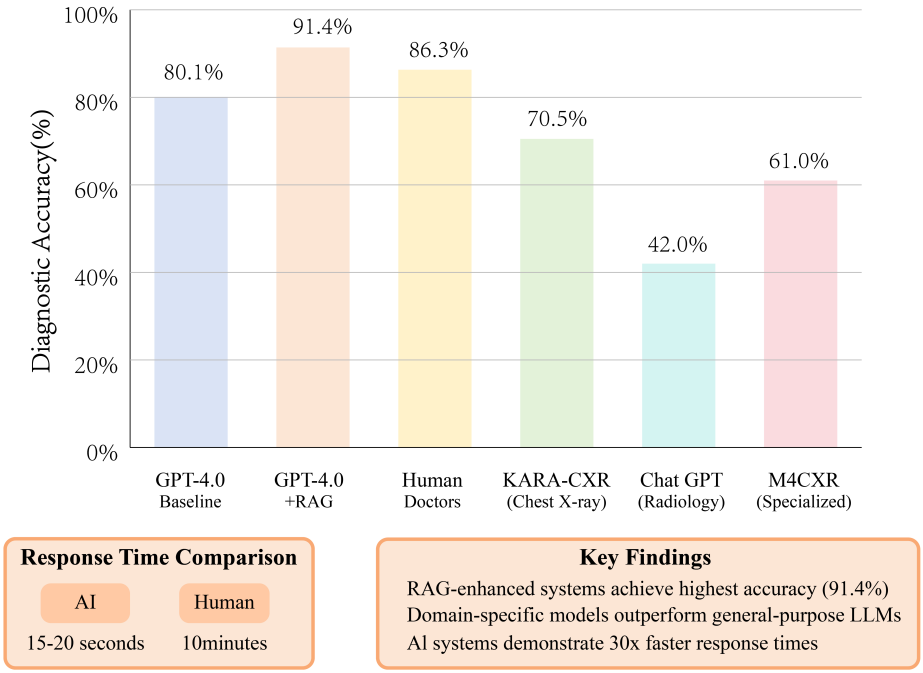



Fig. 5: Comparison of diagnostic accuracy on chest X-ray interpretation: GPT-4.0 baseline, GPT-4.0 with retrieval-augmented generation (RAG), human doctors, KARA-CXR, ChatGPT (Radiology), and M4CXR (specialized). Insets summarize AI vs. human response times and key findings.

assessment approaches for validating the clinical utility of automated medical report generation systems. These studies provide essential insights into the relative performance, strengths, and limitations of AI-driven approaches compared to traditional human-generated medical documentation (As shown in Figure 5).

A comprehensive case study examining LLM-RAG systems in preoperative medicine demonstrated significant performance advantages in both speed and accuracy metrics [112]. Let $T_{\text{human}}$ and $T_{\text{AI}}$ denote the average response times for human professionals and the LLM-RAG model, respectively. The *speedup* factor is given by

$$S = \frac{T_{\text{human}}}{T_{\text{AI}}}, \tag{1}$$

where $T_{\text{human}} \approx 600$ s (10 min) and $T_{\text{AI}} \approx 17.5$ s, yielding $S \approx 34.3$. In terms of accuracy, let $A_{\text{base}}$ be the baseline LLM accuracy and $A_{\text{RAG}}$ the RAG-enhanced accuracy; then the absolute improvement is

$$\Delta A = A_{\text{RAG}} - A_{\text{base}}, \tag{2}$$

with $A_{\text{base}} = 0.801$ and $A_{\text{RAG}} = 0.914$, so $\Delta A = 0.113$. Non-inferiority compared to human accuracy $A_{\text{human}} = 0.863$ was assessed via

$$H_0 : A_{\text{RAG}} - A_{\text{human}} \leq -\delta, \tag{3}$$

rejecting $H_0$ at $p = 0.610$, thus demonstrating non-inferiority.

Specialized medical imaging applications have shown similarly promising comparative results. Research comparing M4CXR, a domain-optimized radiology reporting system, with ChatGPT-4o in chest X-ray interpretation revealed performance differences across acceptability and localization metrics [113]. Let the acceptability rate be

$$R_{\text{acc}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}(\text{report}_i \text{ acceptable}). \tag{4}$$

M4CXR achieved $R_{\text{acc}} \approx 0.61$ versus ChatGPT's $R_{\text{acc}} \approx 0.435$. For anatomical localization accuracy $A_{\text{loc}}$, one observes

$$A_{\text{loc}}(\text{M4CXR}) \approx 0.765, \quad A_{\text{loc}}(\text{ChatGPT}) \approx 0.365.$$

Further validation studies comparing KARA-CXR with ChatGPT for chest X-ray reading demonstrated the impact of domain-specific optimization [114]. Across $N = 2000$ images, KARA-CXR's acceptability ratings $(0.7050, 0.6800)$ outperformed ChatGPT's $(0.4050, 0.4700)$, with interobserver agreement measured by Cohen's kappa:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{5}$$

where $p_o$ is the observed agreement and $p_e$ the expected chance agreement, yielding $\kappa_{\text{KARA}} = 0.74$ and $\kappa_{\text{GPT}} = 0.73$. The non-hallucination rate was $0.75$ for KARA-CXR versus $0.38$ for ChatGPT.

These comparative studies highlight several critical insights:

1) **Domain specificity**: Domain-specific training markedly improves diagnostic metrics.
2) **Efficiency gains**: The speedup $S$ is substantial across applications.
3) **Multi-metric assessment**: Evaluation must encompass accuracy, localization, hallucination rates, and agreement statistics.

### B. Multi-Dimensional Evaluation: Clinical Expert Assessment and NLP Metrics

A comprehensive evaluation of medical report generation systems requires integration of both clinical expert judgment and quantitative NLP metrics to capture the full spectrum of system performance.

Advanced medical QA systems have demonstrated the importance of sophisticated evaluation frameworks that combine multiple assessment dimensions [115]. For an LLM trained on MedQuAD, let precision $P$ and recall $R$ be defined by

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2PR}{P+R}. \tag{6}$$

The Sentence-t5 + Mistral model achieved $P = 0.762$, highlighting the need for precision-oriented metrics.

Systematic evaluation of LLMs in medical evidence summarization has underscored the limitations of automatic metrics and the necessity for human evaluation [116]. Typical automatic metrics such as BLEU and ROUGE-$N$ are given by

$$\mathrm{BLEU} = \mathrm{BP}\,\exp\Big(\sum_{n=1}^{N} w_n \log p_n\Big), \tag{7}$$

$$\text{ROUGE-}N = \frac{\sum_{S\in\text{refs}}\sum_{g\in S}\min(\mathrm{Count}_h(g),\mathrm{Count}_r(g))}{\sum_{S\in\text{refs}}\sum_{g\in S}\mathrm{Count}_r(g)}. \tag{8}$$

However, correlations with human-judged quality are weak, necessitating manual error taxonomy for hallucinations and factual inconsistencies.

The Expert-of-Experts Verification and Alignment (EVAL) framework advances streamlined evaluation for medical LLMs [117]. Define a learned reward model $r_\theta(y|x)$ over candidate answers $y$ given prompt $x$. Models are sampled via rejection sampling: accept $y$ with probability proportional to $r_\theta(y|x)$, improving alignment by rejecting low-score outputs. Fine-tuned ColBERT achieved the highest human alignment, and the reward model replicated human grading in $87.9\%$ of cases, increasing accuracy by $8.36\%$.

Patient-centered evaluation approaches emphasize metrics of patient comprehension. Let $C_{\mathrm{orig}}$ and $C_{\mathrm{pf}}$ be average comprehension scores for original versus patient-friendly reports; the improvement is

$$\Delta C \;=\; C_{\mathrm{pf}} - C_{\mathrm{orig}}\,, \tag{9}$$

with $\Delta C \approx 1.98$ (from 2.71 to 4.69) and observed hallucination and harmful translation rates of $1.12\%$ and $7.40\%$, respectively.

A holistic evaluation metric may be formulated as a weighted combination:

$$\text{Overall Score} = \lambda_1\, A_{\mathrm{clinical}} + \lambda_2\,\mathrm{BLEU} + \lambda_3\,\kappa + \lambda_4\, C_{\mathrm{pf}} \tag{10}$$

The integration of rigorous mathematical definitions with expert insights ensures that medical AI evaluation is both quantitative and clinically meaningful, thereby supporting the safe and effective deployment of LLM-based reporting systems.

## V. Future Directions and Prospects

The future of large language model-based medical report generation is characterized by rapid technological advancement, expanding clinical applications, and growing integration into healthcare workflows. This section examines key development directions that will shape the field, including real-world impact on healthcare delivery, advances in three-dimensional medical imaging, and hybrid approaches that combine multiple AI technologies (As shown in the Figure 6).

### A. Real-World Impact of LLM-Based Medical Report Generation

The integration of large language models into clinical practice is creating transformative changes in healthcare delivery, with implications extending far beyond automated report generation to encompass comprehensive improvements in patient care, clinical efficiency, and medical education [118]. The timeline of LLM development throughout 2023 has demonstrated an unprecedented acceleration in medical applications, with sophisticated models engineered to comprehend, generate, and manipulate medical text on an extensive scale. Healthcare applications of LLMs are evolving from simple text generation tools to comprehensive clinical decision support systems. Current implementations focus on chatbots and interaction systems for clinical documentation management, medical literature summarization, and biomedical natural language processing. However, the next evolutionary leap involves transitioning from AI-powered models primarily designed for answering medical questions to more versatile and practical tools for healthcare providers, including generalist biomedical AI systems for multimodal-based calibrated decision-making processes [118].

Patient engagement and education represent critical areas where LLM-based medical report generation is demonstrating immediate real-world impact. Research on simplified radiology reports has shown that ChatGPT can successfully translate complex medical language into patient-friendly formats [119]. In exploratory studies, radiologists created fictitious radiology reports that were simplified using the prompt "Explain this medical report to a child using simple language." Evaluation by 15 radiologists revealed that most agreed the simplified reports were factually correct, complete, and not potentially harmful to patients. However, instances of incorrect statements, missed relevant medical information, and potentially harmful passages were also identified, highlighting the need for continued refinement and medical expert supervision.

The phenomenon of patients independently using ChatGPT to simplify and explain their medical reports is already affecting patient-doctor interactions, creating both opportunities and challenges for clinical routine. While LLMs demonstrate vast potential to enhance patient-centered care in radiology and other medical domains, realizing this potential while minimizing harm requires supervision by medical experts and continued adaptation to medical field requirements [119]. Integration with next-generation healthcare technologies is creating unprecedented opportunities for comprehensive patient monitoring and care delivery [120]. The fusion of artificial

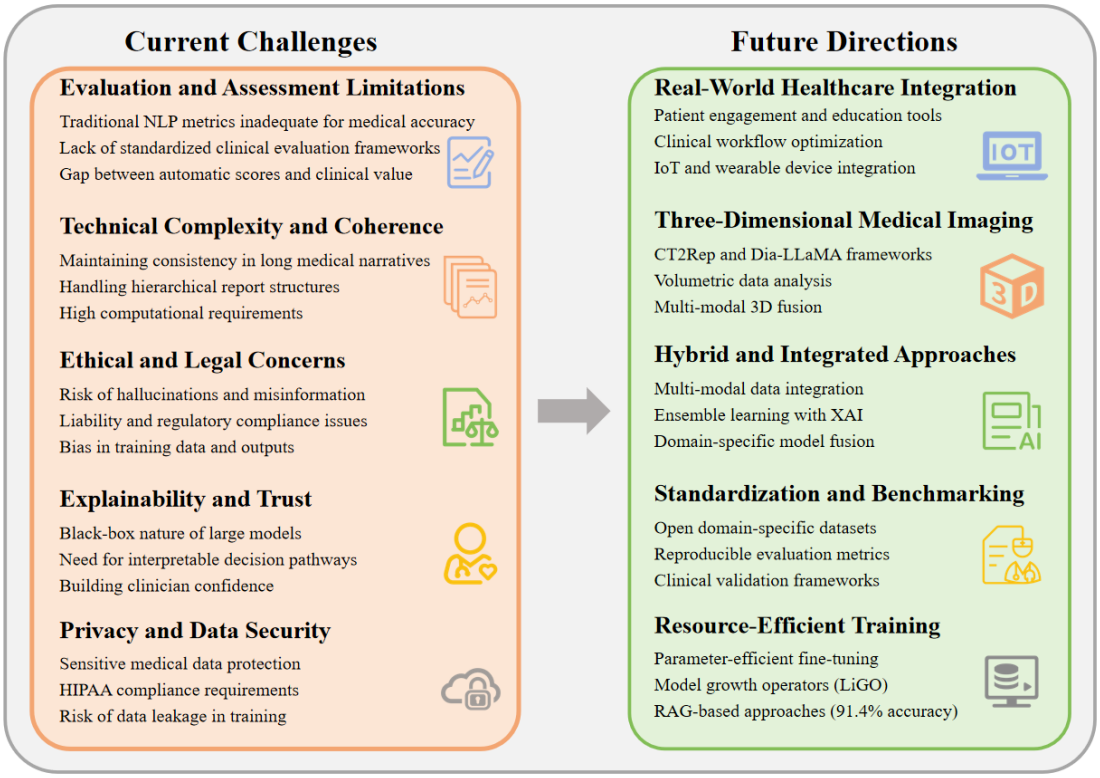



Fig. 6: Current challenges and future directions for medical LLM deployment, covering evaluation and assessment limitations, technical coherence, ethical and legal concerns, explainability, data privacy, real-world integration, three-dimensional imaging, hybrid approaches, standardization, and resource-efficient training.

intelligence with advanced sensors enables intelligent health monitoring systems that provide real-time data collection, online analysis, diagnosis, and treatment recommendations. These closed-loop systems combine Internet of Things technologies, artificial intelligence, and health monitoring sensors to create comprehensive healthcare platforms that extend from wearable sensors for monitoring vital signs to implantable sensors for cardiovascular care and neurotransmitter monitoring. The development of AI-driven information and communication technologies is enabling community-level health monitoring that saves time and allows healthcare providers to serve more patients effectively [121]. AI-based communication ensures reliable interaction, which is crucial in healthcare applications. The integration of AI with Internet of Things technologies creates self-diagnosing capabilities through wireless body area networks using wearable sensing devices, while mHealth technologies enable access to healthcare services anywhere at any time through specialized mobile applications.

### *B. Three-Dimensional Medical Imaging Advances*

The extension of medical report generation to three-dimensional imaging represents a frontier area with significant potential for transforming radiological practice. Traditional approaches have focused primarily on 2D medical imaging, but the computational complexity and data requirements for 3D applications have created unique challenges and opportunities for advanced LLM implementations. CT2Rep represents the first method specifically designed to generate radiology reports for 3D medical imaging, targeting chest CT volumes [122]. This innovative approach addresses the computational complexity and data scarcity challenges that have previously limited 3D medical report generation. The system establishes a baseline using an advanced 3D vision encoder specifically designed for medical imaging, demonstrating effectiveness through a novel auto-regressive causal transformer architecture. The integration of cross-attention-based multi-modal fusion modules and hierarchical memory enables the incorporation of longitudinal multimodal data, leveraging information from previous patient visits to improve diagnostic accuracy and report completeness.

Specialized frameworks for CT report generation, such as Dia-LLaMA, demonstrate the potential for adapting large language models to handle the high-dimensional nature of CT images [13]. This framework adapts LLaMA2-7B for CT report generation by incorporating diagnostic information as guidance prompts, addressing the inherent imbalance in normal and abnormal case distributions that can lead to a biased focus on normal samples. The system leverages pre-trained ViT3D with perceiver architectures to extract visual information and incorporates disease prototype memory banks that are updated during training to capture common disease representations. Disease-aware attention mechanisms enable the model to adjust attention patterns for different diseases, achieving state-

of-the-art performance on both clinical efficacy and natural language generation metrics. Cardiovascular medical imaging represents a particularly promising application area for 3D vision-based medical report generation [123]. Comprehensive surveys of cardiovascular disease analysis based on 3D vision reveal significant advancements in 3D vascular segmentation, 3D vascular map generation, 3D vascular reconstruction, and 3D vascular super-resolution. These techniques provide detailed anatomical information that can significantly enhance the accuracy and clinical utility of automated report generation systems.

The integration of various medical imaging modalities creates opportunities for comprehensive diagnostic systems that can handle diverse data types [124]. Medical imaging modalities, including X-ray, PET scans, CT scans, MRI, and ultrasound, each provide unique information that can be integrated into comprehensive 3D analysis frameworks. The development of intelligent systems using machine learning and deep learning models applied to medical image processing data enables more accurate disease prediction, detection, analysis, and evaluation compared to human interpretation alone. Recent trends in 3D medical imaging emphasize the importance of non-invasive techniques that can provide detailed internal structure visualization without invasive procedures. The integration of medical image processing with machine learning and deep learning models creates intelligent systems that enhance medical assistance and improve recognition accuracy while reducing human error and fatigue in medical image interpretation.

### *C. Hybrid and Integrated Approaches*

The future of medical report generation lies in hybrid approaches that combine multiple AI technologies, integrate diverse data modalities, and leverage the strengths of different computational paradigms to create more robust, accurate, and clinically useful systems. Comprehensive scoping reviews of large language model architectures and applications reveal the rapid evolution toward more sophisticated, multimodal systems [125]. Recent advances encompass architectural improvements, enhanced training strategies, context length enhancements, fine-tuning methodologies, multimodal LLMs, robotics integration, dataset development, benchmarking approaches, and efficiency optimization. The progression from specialized single-task models to versatile, general-purpose systems capable of handling multiple medical domains represents a significant advancement in healthcare AI applications.

Generative AI technologies are creating transformative opportunities for comprehensive healthcare applications [126]. Current implementations span medical imaging, drug discovery, personalized patient treatment, medical simulation and training, clinical trial optimization, mental health support, healthcare operations and research, medical chatbots, and human movement simulation. Real-world healthcare scenarios employing generative AI include visual snow syndrome diagnosis, molecular drug optimization, medical education, and dentistry applications. Healthcare-customized LLMs such as Med-PaLM, BioGPT, and DeepHealth demonstrate the potential for specialized models that can handle domain-specific requirements while maintaining general capabilities.

The adoption of artificial intelligence by healthcare providers depends on multiple factors that must be addressed through comprehensive integration strategies [127]. Research examining healthcare provider behavioral intentions toward AI adoption reveals that performance expectancy, effort expectancy, and initial trust significantly influence adoption decisions. Personal innovativeness positively affects effort expectancy, while task complexity and technology characteristics influence effort expectancy for AI systems. Understanding these factors enables the development of implementation strategies that encourage adoption while addressing potential barriers.

Future developments in radiology report generation emphasize the integration of deep learning with advanced natural language processing techniques [128]. Systematic reviews of deep learning-based research reveal the importance of multimodal approaches that integrate convolutional neural networks and Vision Transformers for medical image feature extraction with transformer-based language models such as BERT, GPT, and T5 for natural language generation. These integrated approaches address challenges including data scarcity, domain-specific variability, and privacy concerns while improving generalizability across different medical imaging modalities.

Comprehensive advancement reviews in medical report generation highlight the evolution from traditional encoder-decoder frameworks to more sophisticated transformer-based models and large language models [129]. Current approaches achieving 92-95% accuracy include encoder-decoder frameworks (45 articles), transformer-based models (20 articles), attention mechanisms (10 articles), RNN-LSTM (10 articles), and graph-based methods (4 articles). However, limitations including overfitting, risk of bias, and high data dependency continue to impact performance and require innovative solutions. Automated radiology report generation represents a rapidly evolving field with significant potential for addressing increasing demands on medical imaging departments [130]. Contemporary approaches emphasize methodological diversity, including contrastive learning, reinforcement learning, CNN and transformer model variations, clinical knowledge integration through multimodal inputs and knowledge graphs, and improved evaluation techniques combining NLP metrics with qualitative clinical reviews.

The future scope of AI in disease detection encompasses ensemble learning combined with explainable artificial intelligence strategies [131]. This approach enhances interpretability, robustness, and clinical relevance by leveraging collective intelligence from multiple algorithms while providing transparent insights into decision-making processes. Future research priorities include the development of hybrid AI architectures, addressing ethical, legal, and societal implications, and leveraging emerging technologies, including the Internet of Medical Things, blockchain, and augmented reality. Multi-modal data integration represents a critical frontier for advancing medical diagnosis through AI technologies [132]. The synthesis of diverse medical data modalities, including images, text, speech, genetic data, and physiological signals enables comprehensive

disease analysis across conditions such as Alzheimer's disease, breast cancer, depression, heart disease, and epilepsy. Advanced feature engineering, classification models, and integration strategies provide valuable insights for clinical decision-making while addressing challenges related to data quality, privacy, and interoperability.

The convergence of these technological advances, integration strategies, and clinical applications suggests a future where LLM-based medical report generation becomes an integral component of comprehensive healthcare delivery systems. Success in this evolution will depend on continued collaboration between AI researchers, healthcare providers, and regulatory bodies to ensure safe, effective, and beneficial implementation of these powerful technologies in clinical practice.

## VI. Discussion and Conclusion

In this review, we have charted the rapid evolution of LLM-based medical report generation, from foundational natural language processing and computer vision techniques to advanced multimodal and retrieval-augmented architectures. Comparative evaluations indicate that retrieval-augmented LLM pipelines can match or exceed junior clinicians' performance in guideline-based tasks, achieving high accuracy and rapid response times. Vision–language models have extended capabilities into volumetric modalities, with pioneering systems for 3D CT report generation and adaptations of large backbone models with disease-aware attention. Despite these advances, several challenges persist. First, traditional NLP metrics such as BLEU and ROUGE-L inadequately capture clinical correctness and completeness, necessitating integrated frameworks that combine automatic scoring with expert clinician evaluations. Second, the black-box nature of large models hinders clinical trust; the integration of explainable AI techniques is essential to illuminate decision pathways, mitigate bias, and support regulatory approval. Third, concerns around patient privacy, data governance, and ethical use demand robust compliance mechanisms and collaborative policy development.

Looking ahead, future research should prioritize:

- **Standardized Benchmarks:** Creation of open, domain-specific datasets for both 2D and 3D imaging and clinical narratives to enable reproducible comparisons across architectures.
- **Explainable and Trustworthy AI:** Development of interpretability methods—such as attention visualization and post-hoc explanations—to foster clinician confidence and facilitate oversight.
- **Hybrid Multimodal Models:** Fusion of structured electronic health record data, genomics, real-time sensor streams, imaging, and text to produce comprehensive, personalized reports.
- **Ethical and Regulatory Frameworks:** Engagement among clinicians, ethicists, and policymakers to establish guidelines for responsible AI deployment, ensuring patient safety and equity.
- **Resource-Efficient Training:** Exploration of parameter-efficient fine-tuning, model growth operators, and retrieval-based strategies to reduce computational costs while maintaining or enhancing performance.

By addressing these priorities, the next generation of LLM-driven medical report-generation systems can achieve greater accuracy, transparency, and clinical utility, ultimately improving patient care and reducing the documentation burden on healthcare providers.

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